

**BANARAS HINDU UNIVERSITY**

B.Sc. (Hons.) — Semester V Examination, 2022-23

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**Paper: GLB-503 — Sedimentary and Metamorphic Petrology**

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 052659

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*Instructions: Answer **five** questions in all, selecting at least **two** each from Sections A and B, and Section-C which is compulsory. All questions are of equal value.*

**SECTION — A**

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**Question 1.** Write an essay on the process of formation of sedimentary rocks. [14]

**Question 2.** Give the classification of sedimentary structures and, with suitable sketches, describe the inorganic primary sedimentary structures forming on the upper bedding planes. [14]

**Question 3.** Write notes on **any two** of the following: [2 × 7 = 14]

- (a) Cement
- (b) Cross-bedding
- (c) Arkoses

**SECTION — B**

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**Question 4.** Discuss in detail the ACF and AKF diagrams and demonstrate their utilities. [14]

**Question 5.** Discuss the phase rule and principles of metamorphic reactions. [14]

**Question 6.** Write notes on **any two** of the following: [2 × 7 = 14]

- (a) Prograde metamorphism
- (b) Metamorphic facies series
- (c) Greenschist facies

**Question 7.** Describe the progressive metamorphism of basic rocks in the CaO – FeO – MgO – Al<sub>2</sub>O<sub>3</sub> – SiO<sub>2</sub> system. [14]

**SECTION — C**

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**Question 8.** Fill in the blanks: [7 × 2 = 14]

- (a) Metamorphism involves the process of \_\_\_\_\_ recrystallization.
- (b) Trace fossils are also known as \_\_\_\_\_.
- (c) Quartzarenite sandstone indicates \_\_\_\_\_ transport.
- (d) In the high grades of metamorphism, \_\_\_\_\_ variety of garnet is present.
- (e) Imbrication is found in \_\_\_\_\_.
- (f) Stromatolites are made up of \_\_\_\_\_ algae.
- (g) Granulite is a \_\_\_\_\_ grade metamorphic rock.